

**TECH
OPS
CRM**

Fit for duty?

IMSAFE - **I**llness · **M**edication ·
Stress · **A**lcohol · **F**atigue · **E**motion.
Any red: hand off.

Every instruction

Three-way: sender states → receiver repeats → sender confirms.

While executing

Sterile cockpit - essential comms only.

Something looks wrong

PACE - **P**robe · **A**lert · **C**hallenge ·
Emergency.
Two unanswered challenges: go over their head.

Major incident

Name roles aloud:
Commander [plan] ·
Comms [external] ·
Ops [keys].
Never one person for all three.

Throughout

Write it down as it happens.

TIME-OUT - STOP, OUT LOUD, EVERYONE

- 1 **Who** Ops on the keys · Monitor watching · who can call the stop
- 2 **What** The change, and the exact system it lands on
- 3 **Rollback** Tested, and how long it takes to run
- 4 **Trigger** The exact condition that means roll back. State it as a number, now.
- 5 **Time-box** When we stop and reassess regardless of progress
- 6 **Gotchas** Anything known about this system that is not in the runbook?
Ask each person by name.

PRE-MORTEM

- 1 The change has failed.
Why? Name three.
Mitigate or accept each, aloud.

GO / NO-GO

Each role answers by name, aloud.

Any No-Go stops it.

Silence is not a Go.

ON THE KEYS

Irreversible action Two-person rule - Monitor confirms before you commit it.

Every command Point and call: read it aloud, Monitor confirms, then execute.

Trigger hit Roll back.
Do not reopen the decision.

SOLO

Irreversible action: there is no solo two-person rule. Defer it, or wake someone.

No Monitor: read the command aloud to yourself, and name the three failures anyway.

RESET - 10 FOR 10

Ten seconds now buys ten minutes.

Stop and ask aloud:

problem? • opinions? • facts? • plan? • distribute?

Trigger it when you feel stuck, not on a timer.

SCOPE - IS / IS-NOT

What, when and where is it failing - and where is it *not*?

The boundary is the clue.

DECIDE - FOR-DEC

Facts • Options • Risks • Decision • Execution • Check

Two options minimum, including doing nothing.

Check it, or it did not happen.

STOP RULES

Fixated	Stand up. Sixty seconds. Then come back.
Past your time-box	Escalate - do not extend. Escalation is not a sin.
Break	Every couple of hours.
Hand off	Before hour twelve - risk more than doubles by then. Past 17 hours awake you are impaired like 0.05% blood alcohol.
Customer waiting	Give a next-update time, then hit it.

SOLO

No Monitor: read the command aloud to yourself before you commit it.

No one to challenge you: ask the pre-mortem question out loud anyway.

SBAR

Situation	What is happening now, and the impact
Background	What changed, and what you already tried
Assessment	What you think it is, and how bad
Recommendation	Exactly what you need from them

CHECK-BACK

They repeat the brief back.
You confirm.

TRANSFER CONTROL

Three exchanges, out loud, in this order:

- 1 "You have control."
- 2 "I have control."
- 3 "You have control."

It is not handed over until all three have been said.

Run this after an incident, **or after any change worth learning from** - including the ones that went fine.

SETUP

- | | |
|-------------------|--|
| Goal | Improve the system, not judge the person. |
| Blame | Shut it down.
Rephrase as: <i>"What made this the reasonable choice at the time?"</i> |
| Data first | Start from the validated timeline.
Ignore speculation. |
| Chronology | Per event:
time [UTC] ·
what the system showed ·
what the human did. |

ANALYSIS

- 5 Whys** Ask why until the answer is systemic.
Never stop at human error.
- SHELL** Cover every dimension:
Software ·
Hardware ·
Environment ·
Liveware [the person] ·
Liveware-Liveware [between people].
- Work-as-done** What did the operator actually do that was not in the runbook?
- Rationale** Ask why it seemed right *then*, not whether it looks right now.
- Defences** List every layer that should have caught this and did not.

ACTIONS

- SMART** Specific · Measurable · Achievable · Relevant · Time-bound.
"Be more careful" is rejected.
- Type** Code/config · documentation · tooling · procedure.
Prefer the last two.
- Owner** One named person, one due date, tracked.
- Verify** Prove the fix works before closing it.

